

VENKATESHA G N

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PROFESSIONAL SUMMARY

Final-year Computer Science / Information Science Engineering student at **RV College of Engineering** (Graduating 2027). Hands-on expertise in full-stack cloud application development, FastAPI/Python, React.js/TypeScript, AWS, Docker, and AI/Vector database integration. Seeking **Full-Time Software Development Engineer (SDE-1)** or Software Engineering Internship roles.

EDUCATION

RV College of Engineering (RVCE)

Bachelor of Engineering in Information Science and Engineering (Final Year)

Bengaluru, India

2024 – 2027 (Expected)

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, System Design Basics.

Government Polytechnic

Diploma in Computer Science and Engineering (Feeder Qualification for Lateral Entry to B.E.)

Chintamani, India

2021 – 2024

TECHNICAL SKILLS

Languages:	Python, Java, TypeScript, JavaScript, SQL, C++
Frameworks & Libs:	FastAPI, React.js, Node.js, Flask, Django, HTML5, CSS3, Tailwind CSS, Bootstrap
Cloud & DevOps:	AWS (EC2, S3), Docker, Docker Compose, Git, GitHub Actions, CI/CD, Nginx
Databases & Vector:	CockroachDB, PostgreSQL, pgvector, MySQL, MongoDB, SQLite, Redis
AI/ML & Security:	LLM API Integration (Gemini), DeepFace, OpenCV, Cryptography (AES-256, RSA)
Core CS:	Data Structures & Algorithms, System Design, OOP, DBMS, Computer Networks, OS

KEY PROJECTS

CareerOS — AI Career Assistant & Persistent Vector Memory Platform

Sep 2026

Tech Stack: Python, FastAPI, React.js, TypeScript, CockroachDB, pgvector, AWS, Docker

github.com/VenkateshGN/careeros

- **Problem Solved & Value Proposition:** Built CareerOS to solve context fragmentation in traditional career platforms (isolated job tools & stateless AI chats), delivering persistent long-term memory assistance across career journeys.
- **Vector Memory & Retrieval Engine:** Architected a semantic vector search engine using **pgvector** and **CockroachDB**, achieving sub-150ms query latency to retain user skills, goals, and resume history across multi-session interactions.
- **Full-Stack Microservices & CI/CD:** Containerized FastAPI backend services and React.js/TypeScript frontend using Docker Compose; automated integration testing via GitHub Actions to reduce deployment friction by 40%.
- **Decoupled AI Intelligence Layer:** Integrated Google Gemini AI via a decoupled service architecture, providing context-aware career recommendations, skill gap analysis, and tailored resume optimization.

Secure Face-Verified E-Voting Management System with Cryptographic Auditing

2025 – 2026

Tech Stack: Python, DeepFace, OpenCV, AES-256, RSA, SHA-256, SQL, Web Technologies

[GitHub Repository](#)

- Engineered a high-security web voting platform featuring 3-factor voter authentication (Credentials, Real-time Biometric Face Verification, OTP) to eliminate proxy voting.
- Integrated OpenCV and DeepFace computer vision pipeline for live webcam verification, achieving <1.5s facial authentication response time.
- Designed an immutable cryptographic voting pipeline utilizing **AES-256-CBC** payload encryption, **RSA digital signatures**, and SHA-256 hash chains for tamper detection.
- Hardened application security against OWASP Top 10 risks including SQL Injection, CSRF, XSS, and privilege escalation attacks.

CERTIFICATIONS

- **NPTEL (IIT/IISc):** Big Data Computing — **Score: 87/100 (Elite)** (Aug – Oct 2025)
- **NPTEL (IIT/IISc):** Data Science for Engineers — Certified (Jan – Mar 2025)

ACHIEVEMENTS & EXTRACURRICULARS

- **Competitive Problem Solving:** Solved 200+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks, focusing on Trees, Graphs, Dynamic Programming, and Backtracking.
- **Engineering Collaboration:** Active participant in technical symposiums and hackathons at RVCE, building scalable software prototypes using modern agile practices.